

RESOURCES FOR CDC

Compiled by: Sarthak Goyal

Many of you will be having your CDC in the upcoming semesters. Summers will be the most important time for your preparation. To help everyone be well prepared for the tests and interview, we have compiled a list of resources for different profiles that you can follow.

This guide is not exhaustive but it will give a good overview of how to prepare for your upcoming CDC. Apart from this, you should also consult your seniors who have got an internship in the respective field you are targeting for a good preparation methodology.

Now let us look at the profiles one by one:

SOFTWARE

To ace the software interview, you need to be good at **Algorithms and Data Structures**. First get the basics of these cleared by following some introductory courses (like CS21003: Algorithms 1 at our college, the NPTEL one [here](#) or any popular course that you can find online.) Then start practising Competitive Programming from the sources listed below or websites like Codeforces, Codechef, Leetcode and Interviewbit.

Sometimes **core CS concepts** are also asked:

- a. **For 3rd years:** Mainly OOPS is asked to 3rd years. Some specific topics in OOPS that have been asked previously are access modifiers, virtual functions, overloading vs overriding, encapsulation, abstraction. Go through <https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/> here or <https://youtube.com/playlist?list=PLlrATfBNZ98dudnM48yfGUldqGD0S4FFb> playlist here
- b. **For 4th years:** 4th years need to mainly focus on OOPS, OS and Networks. A more thorough preparation might include DBMS as well. The portions covered in academics are more than enough to be decently prepared.

More resources for Algorithms and Data Structures Theory

- CP Algorithms website: <https://cp-algorithms.com/>
- CP Handbook:
https://drive.google.com/file/d/1UI_5yAdnDrTtxTcrhwQJ8iOZjFRNR8Nr/view
- Cracking the Coding Interview Book
- Code Readability:
http://personal.denison.edu/~havill/style.pdf?fbclid=IwAR1BtDQGJQBm1z0TYLjyos2sFGdrAAH_9pUelrri5v-1A8cP2RNA9ZE2bFs

Topic Wise Problems For Coding

- Binary Search (fairly imp):
 - <https://www.topcoder.com/community/competitive-programming/tutorials/binary-search>
- STL (very imp)
 - <https://www.topcoder.com/community/competitive-programming/tutorials/power-up-c-with-the-standard-template-library-part-1/>
 - <https://www.topcoder.com/community/competitive-programming/tutorials/power-up-c-with-the-standard-template-library-part-2/>
- Bit manipulation (very imp)
 - <https://www.geeksforgeeks.org/bits-manipulation-important-tactics/>
- Maths (not much asked)
 - https://docs.google.com/document/d/1VVxbGtPR6eNM8AMFv5VYX8S_k1Ksh3VfiihmqavdW5Q/edit?usp=sharing
- DP (very imp)
 - https://docs.google.com/document/d/1MdZlrGHEmIUtrhDDjpguTH-dzytA_MNScCEwPSETHTQ/edit
 - Can refer to <https://cp-algorithms.com/>
 - <https://www.interviewbit.com/> has a good section on DP.
- Graphs (very imp)
 - Can refer to <https://cp-algorithms.com/>
- DP on trees(imp):
 - <https://codeforces.com/blog/entry/20935>

General Practice problems

- Codemonk: <https://www.hackerearth.com/practice/codemonk/>
- A2OJ: <https://a2oj.com/ladders>

- Topic-wise:
 - <https://www.interviewbit.com/>
 - <https://docs.google.com/document/d/1z0Mh7F0dxMVWX2u6WEHWRYqrC4kCXyJ4KEHTkkVYdu8/edit>
- LeetCode Practice:

<https://docs.google.com/spreadsheets/d/1L3cvv0jOKN5hh8GI7iGKG8vmQe81bOi4dBKuFV1uUss/edit#gid=1177592695>

Interview Resources

- Tips for interview:
 - <https://m.facebook.com/ishan.ranga/posts/3264412863648431>
 - https://docs.google.com/document/d/1o4_uJrPaah-rolQxl2G9Uu02Y6kQ4PH7RymENy2pl4/edit
- Experiences:
 - <https://cq-iitkharagpur.medium.com/>
 - https://www.linkedin.com/posts/abhishek8899_internship-interview-experience-abhishek-activity-6691662512074690560-xCqR/
 - <https://codeclub-iitkgp.medium.com/interview-blogs-archive-7849828ed381>
 - https://nitinkgp23.github.io/2020/09/23/A-to-Z-guide-to-your-campus-placements/?fbclid=IwAR3pH6HZ4KCA-DyquhbabRXttGhRH00zlhxo8jg8_Kn-GAp9xpFO5NblmWk
- Some More Resources on Github
 - https://github.com/jnikhilreddy/Competitive-programming-resources?fbclid=IwAR2rJ9ta9eIC-H_HQUB2rKaMb6eJKUJw63nU11CYGlavpM0kN6NDYqoYDOQ
 - https://github.com/Inishan/awesome-competitive-programming?fbclid=IwAR2Wucce017r91pb_hLyOeuQGDhF3i8cVcVAUJtE3tIzP5hxeV-FBdZvTpM

ML / DL / AI / DATA

1. You will first need to be familiar with the basic mathematics behind Machine Learning. This knowledge can be gained through courses in our institute on the topics: **Probability and Statistics** and **Linear Algebra**.
 - a. You can also practise questions from **50 challenging problems in probability**
 - b. If you want to study through video lectures you can follow [Stats 110](#)
 - c. If you are a book person then follow **Prob & Stats by Montgomery**

- d. To solve good dice problems look at <https://www.madandmoononly.com/doctormatt/mathematics/dice1.pdf>
2. Then you can learn the basics of **Python** and important libraries like Numpy, Pandas, Matplotlib
3. Then you can start with **Machine Learning Courses**. Note that you need not do everything here. Just browse through these courses and pick one that you like.
 - a. **Machine Learning by Andrew NG** on [Coursera](#) or [Youtube](#). Make sure you solve the programming exercises.
 - b. **Deep Learning Specialisation** on [Coursera](#)
 - c. The Andrew NG courses though extremely popular are a bit outdated. You can follow [Jovian DL: Zero to GANS](#) and [Jovian ML: Zero to GBMs](#) for a more application-oriented introduction to ML and DL
 - d. For a more theoretical coverage of DL, you can follow **Stanford Courses** like [CS231n](#) for Computer Vision and [CS224n](#) for NLP. Make sure you also visit their course web pages and solve the programming assignments. If you get stuck, you will find their solutions on Github easily.
4. You can move on to programming practice now
 - a. Create an account on [Kaggle](#) and start giving competitions from there.
 - b. Try to do some **hands-on project** following a video from Youtube or some Blog
5. To get **practical experience** in this domain, you can apply for projects under KGP Professors or startups working in this domain.
6. [Machine Learning Tips](#)

QUANT PROFILE

For resources on Quant Profile, you can refer to the following [doc](#) by **Quant Club, IIT Kharagpur**

FINANCE PROFILE

For a Finance profile, it is necessary to have a decent background in Finance on your CV through projects/internships/competitions. Recruiters are generally looking towards finding an inclination towards Finance on your CV while shortlisting your resume, hence a well balanced CV with a decent CGPA increases your chances.

Following is a comprehensive roadmap a beginner can follow to explore the realms of Finance and consider forging a career in the same :-

<https://medium.com/@financeclubiitr/roadmap-to-learning-finance-f3ba8efba298>

In order to learn how to trade and explore the basics of different asset classes, one can read the Zerodha modules and pursue Coursera courses.

<https://zerodha.com/varsity/>

You can also refer to the following books for an extensive introduction and advanced understanding if you have decided to go into depth:-

1. Investments 11th Edition : Bodie, Marcus, Kane (Intermediary)
2. Options, Futures & Other Derivatives : J.C. Hull (Advanced)

As for projects and interns you can apply to various Business Schools through email, Corporates via LinkedIn. Trading, Stock Pitch & Portfolio Management competitions continue to happen all throughout the year and can be found on the Dare2Compete website.

You can also take additional courses from Vinod Gupta School of Management, consider applying for a Masters in Financial Engineering to foster your interest in the field and increase your chances of getting a shortlist.

From an internship/placement point of perspective, most companies have a CV shortlist followed by direct interviews. Some companies would have a quant/coding test to filter in the first round. You are expected to be extremely well versed with every word on your CV, idea about the markets & the area of Finance pertaining to the job description.

CONSULTING

Consulting is about giving solutions to the problems of different companies. This involves data analysis, market research and using your problem-solving skills to solve companies' problems.

To get an idea of consulting you can take part in various case study competitions. In order to explore the field further, you can consider taking internships in corporates/startups in strategy roles, applying for pro-bono consulting projects, etc.

For CDC internships, very few companies come offering a consulting role and follow a strict CV shortlist. In Consulting, CVs are sorted on the basis of 'Spikes'. A spike on your CV is a considerable achievement on a section of your CV (Academics/Project/POR/Extra-Curricular). Spikes on more than one of these sections

is important to secure a shortlist as these roles are open to all departments and secure heavy applications.

Following a shortlist, there are multiple one-on-one case interview rounds followed by a final HR round. Generally firms provide a buddy (recent employee) to help shortlisted candidates prepare for the case interview round. There are multiple resources to practice your case structure, methodology and build a problem solving approach such as :-

1. Case in Point (Marc Cosentino)
2. Case Interviews Cracked (IIT Bombay)
3. Victor Cheng Case Interview Workshops (Youtube)

Following are some resources on Consulting:

1. [Guessing Game: Communique](#)
2. [Previously asked Guestimates](#)
3. [More Resources](#)
4. [Consulting Fundae](#)

PRODUCT MANAGEMENT

A product manager is responsible for the strategy, roadmap, and feature definition for a product or product line. They conduct research, develop strategies, communicate plans and coordinate the development of company projects.

1. Company Wise Mock Interview Videos
 - a. Flipkart: <https://www.youtube.com/watch?v=DSV-vuvmIro&t=148s>
 - b. Microsoft: <https://www.youtube.com/watch?v=Axa7gx1IKqA>
 - c. Google (Maps): <https://youtu.be/CXdaPETE7iE>
 - d. Amazon: <https://www.youtube.com/watch?v=LC92F0nUOV0>
 - e. Lyft: https://www.youtube.com/watch?v=fN6D4i82X_E
2. Interview Tip Videos
 - a. <https://www.youtube.com/watch?v=LVZ-lmQfo4E>
 - b. https://www.youtube.com/watch?v=tpIRd5LKEmY&list=RDCMUcrXxHC_TLVGUy0_zl6E8aNg&index=3
 - c. <https://www.youtube.com/watch?v=61M1r69ZWDg>
 - d. <https://www.youtube.com/watch?v=n530I09t8zY>
3. Community: <http://productmanagementexcercises.com/>

4. Youtube Channels
 - a. PMSchool
 - b. Exponent
5. Topics to be familiar with
 - a. User Journey
 - b. Metrics and KPIs
 - c. Customer Acquisition
 - d. Engagement
 - e. Retention
 - f. Monetization
 - g. Stickiness
 - h. Churn Rate
 - i. Surge Pricing
 - j. 4 Ps of Marketing
 - k. CIRCLES framework
 - l. HEART framework
 - m. Porter's 5 forces
 - n. SEO (learn basic principles)
 - o. How Google Ad and Adwords work
 - p. Have knowledge about Google Analytics
6. Books
 - a. [Decode and Conquer](#)
 - b. [Cracking the PM Interview](#)
 - c. <https://drive.google.com/file/d/1pu6ljOGWIWcWIK1APd8pRt0yC1GbHg97/view?usp=sharing>
7. [Other Resources](#)

SUPPLY CHAIN

Supply chain managers keep track of logistics and update the company's inventory. They analyze operational performance and resolve issues. They also collaborate with vendors and suppliers to ensure all operations (e.g. shipping, delivery) meet quality and safety standards. Their goal is to improve productivity and efficiency and reduce costs while securing high-quality material for their company.

To prepare for a supply chain role, here are some resources

1. Supply Chain Management by Chopra and Meindl
2. Group Discussion Prep Videos on Youtube
3. [More Resources](#)

OTHER IMPORTANT FUNDAMENTALS

1. [Soft Skills Guide by Communique](#)
2. [KGP Life Fundamentals](#)